Vulnerability Assessment Report

1st January 20XX

System Description

The server hardware consists of a powerful CPU processor and 128GB of memory. It runs on the latest version of Linux operating system and hosts a MySQL database management system. It is configured with a stable network connection using IPv4 addresses and interacts with other servers on the network. Security measures include SSL/TLS encrypted connections.

Scope

The scope of this vulnerability assessment relates to the current access controls of the system. The assessment will cover a period of three months, from June 20XX to August 20XX. NIST SP 800-30 Rev. 1 is used to guide the risk analysis of the information system.

Purpose

The database stores a large amount of e-commerce data, including customer information, orders, purchases, and more. This data can be utilized for internal analytics to identify potential customers. The database is critical as it is regularly accessed and used by our employees.

Risk Assessment

Threat source	Threat event	Likelihood	Severity	Risk
Hacker	*Obtain sensitive information via exfiltration*	*3*	*3*	*9*
Employee	Alter/Delete critical information	*2*	*3*	*6*
Power Outage	*Disrupt mission-critical operations*	*1*	*3*	*3*

Approach

The three threats listed in the risk assessment—Hacker, Employee, and Power Outage—were selected based on my assessment of their potential vulnerabilities. A hacker can easily access our database, making it easier for them to launch various attacks or use tools to compromise the data. Employees, on the other hand, could unintentionally execute incorrect queries that may alter or delete critical data. Finally, a power outage could disrupt our business operations, as the database is a crucial asset for our day-to-day activities.

Remediation Strategy

Implementing a strong principle of least privilege and separation of duties is crucial, especially for employees, to prevent misuse of permissions and ensure a clear audit trail. A strong password policy is also necessary to secure the database, especially since it is publicly accessible. Encrypting all sensitive customer data will protect it from hackers or attackers who may attempt to steal it. Lastly, creating database replicas will safeguard against data loss during an outage, ensuring business continuity.